

PSPE: the complete experimental record

CONTENTS

Executive summary

Part I — Method

Part II — Results by module

Part III — The experiments

Part IV — What survives, and what does not

Part V — Measurement defects found and fixed

Part VI — Infrastructure and reproduction

Part VII — Where this leaves the work

Perceive – Simulate – Plan – Explain: constrained intervention design for PDE-governed systems. Consolidated results, updated 2026-09-24.

Every number here is five seeds unless stated otherwise, reported as mean ± sample standard deviation, with paired t where two arms are compared ($df = 4$ for five seeds, so $|t| > 2.78$ is $p < 0.05$; $df = 2$ for three seeds, so $|t| > 4.30$). Each section names the run directory its numbers come from. Detailed analyses live in `docs/results/`; this document is the single place where all of it is stated together, including the claims that did not hold.

Executive summary

PSPE is four modules trained around one loop: turn an observation of a physical hazard into a state estimate, roll a differentiable surrogate forward, choose an intervention under hard constraints, and write a brief a human can act on. This record covers everything measured on it.

Four contributions, as they stand after all testing:

#	contribution	status
1	Safety by measurement. Model-based constrained planning fails silently because the dual controller trusts the surrogate's cost. Probing the real system and planning against a measured margin fixes most of it.	Holds. Two PDE families. Violations cut ~7x on dar (1.8% to 0%) and 85% to 5.5% on rdf. The same diagnosis repaired PPO-Lagrangian.
2	Sample efficiency. The planner acts inside the surrogate, so its real-world cost is the data the surrogate was fitted on.	Holds. 9x fewer real transitions than model-free safe RL on dar, 5x on rdf, at comparable return.
3	Planning on observed data. Constrained firebreak planning through a surrogate fitted to real fire spread.	Holds, 5 seeds. +12.0 points of burn reduction over the operational forecast-then-greedy heuristic at identical budget, zero violations, paired $t = 11.15$. Advantage is 1.9–2.1x at the tightest budgets, widens when the imposed action physics is removed, and grows monotonically with horizon ($t = 1.40$ at 1 day, +14.3 at 5 days) — the mechanism, measured. Constrained-RL baselines on the same fires reach 4.6%, below the heuristic's 24.1%.
4	Certified explanations. Briefs trained against the policy, with a conformal bound on their faithfulness.	Half withdrawn. The certificate's coverage holds on real data. The faithfulness claim does not: a permutation control shows the briefs carry no state-specific information.

The three findings that changed the project:

1. The planner's constraint violations came from controlling the dual on surrogate cost while violation is realised under true dynamics. Measuring reality periodically and planning against a margin is the fix, and it is contribution 1.

- 2. Joint Simulate+Plan training — the paper's original headline — never changed a decision on any of three PDE families. Anchored to data it improves the surrogate 5–9x; unanchored it destroys it. A clean negative with a safety lesson.
- 3. Explanation faithfulness was never being measured. A permutation control (score each brief against a *different* state's action) shows zero gap on every arm and both testbeds, and the synthetic testbed used for the published numbers has a state-independent policy, so it could not have detected the difference.

Part I — Method

Four modules, one loop.

Perceive. A frozen vision backbone (SigLIP) with LoRA adapters and two heads: a decoder that reconstructs the physical field, and a projection head aligned to caption embeddings by InfoNCE. Loss $L = w_{reg} L_{reg} + w_{con} L_{con}$.

Simulate. A Fourier Neural Operator: lift, then blocks combining a spectral path (FFT, keep modes $k \leq 12$, learned complex weights, inverse FFT) with a pointwise 1x1 convolution, then project. Trained on one-step, rollout and physics-residual terms. Optionally spectrally normalised so its Lipschitz constant is bounded.

Plan. A tanh-squashed Gaussian policy over actuator amplitudes. The Lagrangian $-(r - \lambda c)$ is differentiated two ways — pathwise through the surrogate (low variance, biased by surrogate error) and likelihood-ratio (unbiased, high variance) — and mixed at the variance-minimising weight. A PID-Lagrangian dual drives λ from measured cost. Constraints: intervention budget, pointwise safety, and a distributional equity term.

Explain. A frozen small LM (Qwen2.5-0.5B) with LoRA, conditioned on a prefix built from the planner's state and action. A frozen parser reads the generated brief back into an action distribution, and $F(b) = \exp(-KL(\pi || \text{parse}(b)))$ scores the agreement. A split-conformal certificate bounds F on unseen states.

Testbeds. Three PDE families — `dar` (diffusion–advection–reaction), `swe` (shallow water), `rdf` (reaction–diffusion front) — plus two real datasets: PDEBench 2D shallow water, and Next Day Wildfire Spread (NDWS).

Part II — Results by module

Perceive

Real SigLIP backbone (`google/siglip-base-patch16-224`), 3 seeds (`runs/perception_real/`):

arm	field rel L2	retrieval accuracy	trainable
PSPE — frozen + LoRA + contrastive	0.0596 ± 0.027	0.754 ± 0.026	1.64 M
probe — frozen backbone, decoder only	0.0345 ± 0.011	0.135 ± 0.009	1.64 M
CNN from scratch	0.0564 ± 0.015	0.130 ± 0.016	0.56 M

The frozen probe reconstructs best and beats the CNN ($t = -4.87$). The design wins on alignment instead: retrieval 0.754 vs 0.130, $t = +37$. Real SigLIP features are strong enough that adapters do not help reconstruction. The stub backbone gave the same shape (0.0524 / 0.0476 / 0.0422; retrieval 0.510 vs 0.137).

Reading. PSPE's perception trades reconstruction accuracy for text–image alignment. It does not beat a plain CNN at estimating the field.

Simulate

PDEBench 2D shallow water, 512 real trajectories, 64^2 , 20 epochs (`runs/simulate/`, `runs/pdebench/`):

operator	rel L2 (1-step)	rel L2 (rollout)	params	published FNO (nRMSE)
FNO	0.0018	0.0098	1.19 M	0.0044
DeepONet	0.0044	0.0137	0.09 M	—
GNOT	0.0056	0.0543	0.31 M	—

Metrics differ (relative L2 vs nRMSE), so this is an anchor, not a like-for-like comparison. The operator ranking reproduces on synthetic data (FNO 0.057, DeepONet 0.256, GNOT 0.268, 5 seeds), which is the more robust claim.

Resolution invariance (runs/resolution_seeds/), one FNO trained at 64^2 and evaluated at three grids, 5 seeds:

grid	cells	rel L2
64^2	4,096	0.0298 ± 0.014
96^2	9,216	0.0298 ± 0.014
128^2	16,384	0.0300 ± 0.014

Flat to three decimals: genuine discretization invariance.

Observed wildfire spread (runs/ndws_seeds/), ~18k real 64 km patches, 1.3% fire prevalence, AUC-PR on the validation split, 3 seeds:

model	AUC-PR	vs published 0.284
U-Net	0.277 ± 0.002	97.5%
Hybrid (U-Net + gated FNO)	0.235 ± 0.005	83%
FNO alone (1 seed)	0.200	70%
all-zeros floor	0.011	—

The U-Net beats the hybrid on every seed ($t = +12.5$), and the hybrid's learned gate over its spectral path — free to take any value, initialised at zero — settles at **0.12 on every seed**. Offered the Fourier operator, the model declines it.

On the test split (runs/ndws_test/ , 3 seeds, the split the published 0.284 is reported on, with flip augmentation):

model	test AUC-PR	vs published 0.284	spectral gate
U-Net	0.3162 ± 0.017	111%	—
Hybrid (U-Net + gated FNO)	0.2939 ± 0.003	103%	0.111

The surrogate **exceeds** the published benchmark on its own split. Two conditions belong with that number: training used 8,000 patches rather than the full train split, and this run used flip augmentation (with the wind-direction channel corrected under the flip) which the validation runs did not. The spectral gate settles at 0.111 again, a third independent replication of the operator-scope finding.

Reading. The operator's scope, measured: excellent on smooth PDE fields and resolution-transferable; loses to convolutions on sharp fronts, because a spectral model truncates the high frequencies that *are* a fire front. PSPE survives this because it is a loop with a swappable surrogate.

Plan

Against safe RL on dar (runs/seeds_v3/), 5 seeds, cost limit 0.936:

method	return	cost	violating evals	worst	real samples
PSPE (adaptive α)	-2.448 $\pm$ 0.029	0.412 $\pm$ 0.37	7.3%	1.109	3,072
PSPE (fixed α)	-2.738 $\pm$ 0.365	2.752 $\pm$ 3.6	12.7%	2.758	3,072
PPO-Lagrangian	-2.558 $\pm$ 0.039	0.266 $\pm$ 0.009	0%	0.297	38,400
CPO	-2.541 $\pm$ 0.035	0.263 $\pm$ 0.011	0%	0.277	38,400
Sauté RL	-2.537 $\pm$ 0.024	0.264 $\pm$ 0.008	0%	0.275	38,400
Primal-dual NPG	-2.544 $\pm$ 0.026	0.263 $\pm$ 0.008	0%	0.275	38,400

Beats every baseline on return ($t = +3.84$ to $+17.95$) at 12.5x fewer real transitions — but violates a limit they all respect. That failure and its fix are Part III.

Against safe RL on rdf (`runs/baselines_rdf/` , after the planner fix):

method	return	violating evals	real samples
PPO-Lagrangian (stock gains)	-7.66 $\pm$ 0.31	1.8%	38,400
PPO-Lagrangian (with our dual fix)	-7.69 $\pm$ 0.31	0%	38,400
CPO / Sauté / NPG	-7.97 to -8.08	0%	38,400
PSPE v2 (probe + margin)	-7.49 $\pm$ 0.62	5.5%	7,360

Paired t vs PPO-Lagrangian: **+0.82**. Parity on return, 5.2x fewer real samples, and PPO-Lag has the better tail. CPO, Sauté and NPG are safe by inaction: their cost sits at the calibrated do-nothing level.

Reading. The return edge is a dar result, not a general one. Sample efficiency and the safety mechanism hold on both families.

Explain

The full Explain story is Part III §5–6. In summary: the conformal certificate's coverage holds on real data, and every claim about the briefs being faithful to a particular decision has been withdrawn.

Part III — The experiments

1. The constraint failure, and the fix

The failure. The planner exceeded the cost limit on 7.3% of evaluations (worst case 1.109 against 0.936) while every model-free baseline exceeded it on none. The mechanism is a property of model-based planning, not a coding error: the dual controller is updated from cost measured on **surrogate** rollouts, while violation is realised under the **true** dynamics. When the surrogate under-predicts cost, the policy is safe in-model and unsafe in reality.

The fix. Two mechanisms (`runs/constraint_fix/`), 5 seeds:

- a **probe** that rolls the current policy in the true environment every 20 iterations and feeds that cost to the dual, with the surrogate's cost bias-corrected in between;
- a **margin** that plans against `limit - k  $\sigma$` , σ being the measured spread of surrogate cost error.

arm	return	violating evals	worst cost	real samples
baseline	-2.302 $\pm$ 0.16	1.8%	0.75 (max 1.73)	3,200
probe only	-2.307 $\pm$ 0.16	1.8%	0.56	4,160
margin only	-2.302 $\pm$ 0.16	1.8%	0.75	3,200
probe + margin	-2.301 $\pm$ 0.17	0% on all 5 seeds	0.29	4,160

Neither half works alone. The margin alone is *identical* to the baseline, because with no probe there is no measured error to derive a margin from: **conservatism needs a measurement**.

Honest rate. Across all 15 seed-runs that use probe + margin (the constraint-fix, alpha-rule and joint runs), 13 are clean and two had a single evaluation over the limit. So the headline is a ~7x reduction — about 1% of evaluations versus 7.3% — not a guarantee.

Attempting a guarantee, and getting the quantity wrong. A split-conformal margin should convert that reduction into a stated failure rate: with n probe observations and level δ , the $\text{ceil}((n+1)(1-\delta))$ -th order statistic is a margin under which $P(\text{cost} > \text{limit}) \leq \delta$. The first implementation conformalised the **surrogate's cost error** and made rdf worse — 18.2% violating against the 14.5% baseline, with a margin of 0.25 where the k-sigma rule had found 0.69:

testbed	arm	violating evals	effective limit
dar	probe + conformal margin	0%	0.769
rdf	baseline	14.5%	3.260
rdf	probe + conformal margin (over model error)	18.2%	3.010

The diagnosis is that the bound was over the wrong distribution. On rdf the surrogate is accurate (cost bias -0.08); what violates the limit is the **policy's own episode-to-episode spread**, which a quantile over model error never sees.

Conformalising the right quantity — the deviations of individual real episode costs from their probe mean, plus any systematic surrogate bias — gives a margin that holds (`runs/constraint_fix_conf*`, 5 seeds, $\delta = 0.1$, probe every 10 iterations):

testbed	arm	violating evals	target	worst cost	return	effective limit
dar	baseline	0%	—	0.225	-2.304	0.936
dar	probe + conformal margin	0%	$\leq 10\%$	0.225	-2.309	0.676
rdf	baseline	14.5%	—	4.00	-7.184	3.260
rdf	probe + conformal margin	7.3%	$\leq 10\%$	3.51	-7.351	2.777

The bound holds on both families. Against the hand-tuned k-sigma margin it replaces (5.5% violating at return -7.49 on rdf), the conformal version gives slightly more violations but a **stated failure rate** rather than a tuned constant, and costs 0.14 less return. This is the form the claim should take: not "violations fell 7x", but " $P(\text{cost} > \text{limit}) \leq \delta$, and at $\delta = 0.1$ the realised rate was 7.3% and 0%".

The failure is worth keeping: a margin is only as good as the distribution it bounds, and conformalising a conveniently available quantity rather than the one that violates gives a bound over the wrong thing.

2. The planner on a second family: rdf

As shipped, the planner violated the rdf limit on **85%** of evaluations at 2.5x the limit, in every arm, regardless of the probe. Three defects, each read off the training trace once the previous one was removed

(`docs/results/RDF_PLANNER_FIX.md`):

Defect 1 — action saturation. The reward-greedy sprint drives the pre-tanh policy mean past $|\mu| \approx 3$, where tanh has no slope:

iteration	cost	λ	α	llg_pathwisell	llg_LRll	var_pathwise
20	1.05	0	0.70	0.55	2.9	1e-2
40	5.87	0	0.81	1.9	196	0.7
100	8.08	14.5	0.95	0.036	5,160	0.0
198	8.21	38.5	0.99	0.036	12,700	0.0

The pathwise gradient dies (norm 0.036, variance exactly zero); the variance-optimal mixing rule then *picks the dead branch* ($\alpha \rightarrow 0.99$), because zero variance looks like precision; and the likelihood-ratio branch explodes because every rollout returns the same value and the advantage normaliser divides by ~ 0 . $\lambda = 38$ multiplies nothing. Fix: a soft wall $c \cdot \text{relu}(|\mu| - 1.5)^2$ on both branches and a floor on the advantage std. **85% $\rightarrow$ 44%**.

Defect 2 — dual oscillation. λ collapses to zero whenever cost dips under the limit; the policy sprints greedy again and overshoots. Lowering the gain made it *worse* (60%). Raising the integral gain $\times 10$ lets λ hold. **44% $\rightarrow$ 14.5%**.

Defect 3 — tail violations. A dual that holds the *mean* at the limit violates on about half of evaluations by construction. The margin now also covers the policy's own episode-to-episode spread, measured at each probe. **14.5% $\rightarrow$ 5.5%**, worst case 3.23 against a 3.26 limit.

dar regression check. The same configuration on dar: return unchanged (-2.304 vs -2.302), 0% violations on every arm, worst case 3x smaller. One planner configuration now works on both families.

3. Joint Simulate+Plan training

The paper's original contribution 1. One pretrained surrogate, deep-copied per arm, 5 seeds (`runs/joint/`, `runs/joint_swe/`, `runs/joint_rdf/`):

testbed	arm	return	held-out surrogate rel L2 before $\rightarrow$ after
dar	disaggregated	-2.302 $\pm$ 0.17	0.0048 $\rightarrow$ 0.0048
dar	joint, anchored	-2.296 $\pm$ 0.18	0.0048 $\rightarrow$ 0.0005
dar	joint, unanchored	-2.369 $\pm$ 0.17	0.0048 $\rightarrow$ 0.975
swe	disaggregated	-0.1733 $\pm$ 0.010	0.0246 $\rightarrow$ 0.0246
swe	joint, anchored	-0.1733 $\pm$ 0.010	0.0246 $\rightarrow$ 0.0027
swe	joint, unanchored	-0.1733 $\pm$ 0.010	0.0246 $\rightarrow$ 1.14
rdf	disaggregated	-6.76 $\pm$ 0.69	0.0024 $\rightarrow$ 0.0024
rdf	joint, anchored	-6.70 $\pm$ 0.70	0.0024 $\rightarrow$ 0.0005
rdf	joint, unanchored	-7.46 $\pm$ 0.45	0.0024 $\rightarrow$ 0.83

Paired t on return, joint vs disaggregated: **+0.66** (dar), identical (swe), **+1.00** (rdf). Unanchored vs disaggregated: -5.84 (dar), -2.41 (rdf).

Reading. Joint training never changes the decision. Anchored to data it improves the surrogate 5–9x on held-out trajectories — the planning gradient, scaled to a tenth of the data gradient, acts as extra training on the states the policy visits. Unanchored it destroys the surrogate. The claim "joint beats disaggregated" is not supported on any testbed; "joint is safe when anchored and destructive when not" is supported on all three.

3b. swe: a testbed that could not rank methods, and why

`runs/calibration/swe_u*.json` plus direct probes. The calibration procedure checked that the **cost** separates between doing nothing and a reward-greedy policy, so the constraint binds. It never checked that the **return** separates, so actuation can move the objective. swe passed the first and failed the second for the whole project.

probe	result
400 random constant action vectors	not one beats doing nothing
hand-written state-feedback damper	+0.0098 (5.5% of the objective)
trained reward-greedy planner	+0.0122 (7% of the objective)

So the task is controllable and the planner does find control — it beats both doing nothing and the hand damper. The achievable band is simply $\sim 7\%$, and the differences between planning methods live inside it. That is why swe's joint / disaggregated / unanchored arms all returned -0.1733 to four decimals.

A correction to earlier drafts of this report, which said the swe policy never leaves its initial behaviour: it does, and it improves. The band is too narrow to rank methods within, which is a different and more precise statement.

The fix that does not work. Raising actuator amplitude grows the margin for a *deterministic* controller (5.5% at amplitude 1.0 → 32.6% at 4.0) but gives a stochastic policy nothing, because amplitude scales the useful signal and the exploration noise together: planner separation is +0.0122 at amplitude 1.0 and +0.0081 at 4.0. Amplitude was left at 1.0 and `return_separation` is now a first-class output of the calibration script.

What would work: an objective whose optimum is far from zero action — tracking a non-trivial target profile — rather than driving height to zero in a system where localised forcing only injects energy.

4. Theory checks

Assumption 1 and Proposition 1 (`runs/lipschitz/`), 5 seeds:

arm	rel L2	L_G	L_G ≤ 1	measured return bias	Prop 1 bound, ∞-horizon	bound at H = 12
default FNO	0.069 ± 0.062	0.985 ± 0.013	4 of 5	0.10	422	9.8
spectrally normalised	0.079 ± 0.026	0.964 ± 0.003	5 of 5	0.11	748	17.3

Assumption 1 holds after training (initial L_G was 1.61). The infinite-horizon bound is **vacuous** — 422 against a measured bias of 0.10, because the $1/(1-\gamma)$ prefactor at $\gamma = 0.98$ is 2,450. The finite-horizon form holds on every seed and is still 100x loose. Report the finite-horizon form and say it is loose.

Equation 8 mixing rule (`runs/alpha_rule/`), 5 seeds:

arm	return	violating evals	final α	measured B ²	real samples
fixed $\alpha = 0.5$	-2.415 ± 0.30	3.6%	0.5	—	4,160
variance rule	-2.302 ± 0.17	1.8%	0.992	—	4,160
Eq. 8 (measured bias)	-2.299 ± 0.15	0%	0.992	0.0013	6,080

The measured pathwise bias is ~0.001, so Eq. 8 and the variance-only rule choose the same α . Return $t = +0.22$. Eq. 8's zero-violation result comes from its 1,920 extra truth rollouts, not from the formula.

Adaptive α is a variance effect, not a mean effect: +0.290 return over fixed α but $t = +1.66$. What separates them is the tail — std 0.029 vs 0.365, and one fixed- α seed diverging to cost 8.1 with $\lambda = 19.1$.

Cross-family transfer (`runs/transfer_seeds/`), 5 seeds:

transfer	fidelity gap	measurable?
rdf → dar	0.358 ± 0.033	yes, ~11 σ from zero
swe → rdf	0.311 ± 0.13	yes
dar → rdf	1.81 ± 1.1	marginal
swe → dar	0.087 ± 0.14	no, crosses zero
rdf → swe	10.5 ± 8.2	no, $\sigma \approx$ mean
dar → swe	27.4 ± 26	no, $\sigma \approx$ mean

Direction survives (parabolic families transfer; transfer into the wave family collapses); magnitude does not. And forecast error decouples from decision loss: dar → swe has a fidelity gap of 27 and a planning gap of 0.007 ± 0.014 (nothing), while dar → rdf has fidelity gap 1.8 and planning gap 0.58 ± 0.22 (about 8% of return).

Ablations (5 seeds, full budget):

component	on	off	t	verdict
physics-informed loss	0.0567 ± 0.026	0.0496 ± 0.044	+0.52	no effect on final accuracy
frozen perception	0.0681 ± 0.013	0.0908 ± 0.017	-3.97	frozen wins
faithfulness term	0.643 ± 0.15	0.598 ± 0.006	+0.67	no effect

The physics loss buys convergence speed only (final rel L2 0.117 with it vs 0.273 without at a quick budget).

5. Planning on observed wildfire data

The first planning result whose dynamics are a model of what a real fire did (docs/results/NDWS_PLANNING.md). The NDWS U-Net, frozen, with an 8×8 grid of firebreak intensities per day under a 3%-of-patch daily crew budget, three days ahead, population-weighted burn as the objective. Action model, stated in code: a break removes fuel ($\text{NDVI} - 2\sigma \times \text{intensity}$, the learned model decides the effect) and blocks spread into treated cells ($0.9 \times \text{intensity}$, imposed). 1,500 held-out fires, 3 seeds:

policy	burn reduction	treated/day	over budget	via learned fuel channel
random	$4.5 \pm 0.3\%$	3.0%	0	0.6%
greedy: forecast today, treat riskiest cells, repeat daily	$24.1 \pm 3.8\%$	3.0%	0	5.7%
PSPE per-instance: plan three days jointly through the surrogate	$36.5 \pm 1.0\%$	3.0%	0	19.3%
PSPE amortised policy	$12.2 \pm 4.5\%$	2.2%	17%	2.5%
unconstrained	$85.7 \pm 4.5\%$	61.7%	100%	24.4%

Per-instance minus greedy, per seed: +9.6, +11.7, +15.9 points. Paired t (df 2) = **6.66**.

Extended to 5 seeds (runs/ndws_plan/ , seeds 3–4 added):

policy	burn reduction, 5 seeds	per-seed
greedy	$22.0 \pm 4.4\%$	28.0, 23.9, 20.4, 21.9, 15.9
PSPE per-instance	$34.0 \pm 4.4\%$	37.6, 35.6, 36.3, 34.1, 26.5
PSPE amortised	$10.4 \pm 6.5\%$	—

Paired t (df 4) = **11.15**, mean gap +12.0 points. The two added seeds did not dilute the effect: the paired gap is stable from +9.6 to +15.9 across all five.

5a. The crew budget sweep (Pareto)

Same fires and seeds, budget varied (runs/ndws_plan_b*/):

crew budget / day	greedy	PSPE per-instance	ratio
1%	$7.5 \pm 1.4\%$	$14.3 \pm 0.5\%$	1.91x
2%	$12.8 \pm 2.0\%$	$26.2 \pm 0.7\%$	2.05x
3%	$24.1 \pm 3.8\%$	$36.5 \pm 1.0\%$	1.52x
5%	$36.1 \pm 5.0\%$	$53.0 \pm 1.1\%$	1.47x
8%	$55.0 \pm 5.5\%$	$69.4 \pm 0.6\%$	1.26x

The advantage is **largest where crews are scarcest** and decays as the budget grows large enough that placement stops mattering. PSPE at a 1% budget matches greedy at close to 2%, i.e. roughly half the crews for the same outcome. The planner's variance is also consistently smaller (± 0.5 to ± 1.1 against ± 1.4 to ± 5.5): it is not only better on average but steadier across fires.

5a2. Constrained RL on the same fires

`runs/ndws_baselines_fair/`, 5 seeds. Every arm plans or learns against the same frozen U-Net under the same 3% daily budget and is scored by the same function on the same held-out fires. Sample efficiency does **not** apply here: there is no separate true environment, so every method draws from the same surrogate and this is a comparison of decision quality only.

method	burn reduction @ matched budget	treated/day	fires over budget
greedy heuristic	24.1 ± 3.8%	3.00%	0
PPO-Lagrangian	4.6 ± 0.4%	4.32%	58%
CPO	4.3 ± 0.1%	2.84%	6%
Sauté RL	4.7 ± 0.5%	7.37%	79%
primal-dual NPG	4.6 ± 0.1%	3.26%	100%
PSPE per-instance	36.5 ± 1.0%	3.00%	0

The constrained-RL methods land near random placement (4.5%) and below the heuristic. The reason is amortisation, not the constraint: one policy has to serve 1,024 distinct fires from 200 iterations, which is the same failure our own amortised arm shows (10.4%). On a per-instance decision problem this diverse, decision-time optimisation is the right tool and a learned policy is not.

A near-miss worth recording. The first version of this comparison had all four baselines treating 22–38% of the sector against the 3% budget on every fire, which looked like a structural finding about constrained RL in high-dimensional action spaces. It was an artefact: under the squared intensity map a randomly initialised policy emits 25% treatment, and the agents had barely moved off initialisation, while PSPE's optimiser *starts* at the budget and projects onto it every step. With `--fair-init` giving the baselines the same starting point, PPO-Lagrangian holds 2.99% after three iterations. The structural claim was withdrawn before it was made. See `docs/results/ROBUSTNESS_NOTES.md`.

5a3. Horizon ablation: planning pays only when the decision is sequential

`runs/ndws_horiz_h*/`, 5 seeds per horizon, budget 3%/day:

horizon	greedy	PSPE per-instance	gap	paired t (df 4)
1 day	24.8 ± 1.1%	25.4 ± 1.7%	+0.6	1.40 — not significant
2 days	24.2 ± 3.6%	32.6 ± 3.4%	+8.5	10.74
3 days	22.0 ± 4.4%	34.1 ± 4.4%	+12.1	11.08
5 days	17.7 ± 4.3%	32.0 ± 5.8%	+14.3	8.99

(Significance threshold at df 4 is $|t| > 2.78$.)

At a one-day horizon the two are statistically indistinguishable ($t = 1.40$), which is the prediction, not a disappointment: the one-day objective is separable over cells, so treating the highest-risk cells within budget is its exact optimum and no planner can beat it. The moment the decision becomes sequential the statistic jumps by an order of magnitude ($t = 10.74$ at two days), and the gap grows monotonically with horizon while the heuristic **degrades** (24.8% → 17.7%) because it keeps optimising for today while the fire moves.

A result that is insignificant exactly where theory says it must be, and significant exactly where theory says it should be, is stronger evidence for the mechanism than any single large number.

This is the mechanism behind the headline result, measured rather than asserted: planning buys nothing on a myopic problem, and its value grows with how far ahead the decision reaches.

5b. Fuel-only ablation: does the result rest on the imposed physics?

The action model has two mechanisms — a **learned** one (fuel removed from the NDVI channel, the surrogate decides the effect) and an **imposed** one (spread into treated cells suppressed by hand). Setting `block = 0` removes the imposed half entirely (`runs/ndws_plan_fuelonly/` , 3 seeds, budget 3%):

	greedy	PSPE per-instance	gap
both mechanisms	24.1 ± 3.8%	36.5 ± 1.0%	+12.4 (t = 6.66)
learned fuel response only	5.7 ± 2.9%	19.4 ± 1.6%	+13.6 (t = 5.93)

Removing the hand-written physics costs the heuristic 76% of its effect and the planner 47%, and the **gap widens**. The planner's advantage therefore does not rest on the part of the action model that was specified rather than learned — it comes from exploiting the surrogate's own fuel response, which is what the claim requires. This is the ablation a reviewer asks for, and it survives.

Reading. Decision-time planning beats the operational heuristic by twelve points at the same budget with no violations, and the margin comes through the learned model: 19% of the planner's reduction survives with the imposed block term switched off, against 6% for greedy. Greedy treats today's risk; the planner treats where the model says the fire will be on day three. On the one-day version of the problem greedy is the *exact* optimum of a separable objective and nothing beats it — planning earns its keep only across days. The amortised policy is the wrong tool on a distribution this diverse.

Not shown: any effect on a real fire. NDWS records what fires did without firebreaks; there are no counterfactuals. The forecast is validated against observation (AUC-PR 0.264–0.280 on these same held-out fires); the treatment effect rests on the stated action model.

6. Explanation: the certificate holds, the faithfulness claim does not

What was claimed. Trained-in briefs beat a genuine post-hoc control: on dar with Qwen, 0.468 ± 0.16 vs 0.182 ± 0.16 , $t = +21.3$; on real fire plans, 0.814 ± 0.039 vs 0.741 ± 0.005 , $t = +3.61$. A split-conformal certificate holds at 90% and 95% coverage on 3 of 3 seeds, with a non-vacuous floor ($F \geq 0.706$ at $\delta = 0.1$ on real fire plans).

The control that broke it. Score each brief against a **different** state's action distribution. If briefs are state-specific, aligned F should exceed shuffled F. If the generator emits the same brief regardless of state, the two are equal by construction:

testbed	arm	F aligned	F shuffled	gap
dar, seed 0	trained-in	0.2809	0.2805	+0.0004
dar, seed 0	post-hoc	0.0034	0.0034	0.0000
dar, seed 1	trained-in	0.5570	0.5570	0.0000
dar, seed 2	trained-in	0.5657	0.5659	-0.0002
NDWS, seed 0	trained-in	0.7439	0.7436	+0.0003
NDWS, seed 1	trained-in	0.7673	0.7674	-0.0001
NDWS, seed 2	trained-in	0.8630	0.8639	-0.0009

Zero state-specific information, on both testbeds, for every arm including the post-hoc control. The sampled briefs confirm it: the same actuators at the same amplitudes for different states, with only the cost and reward lines varying.

Three fixes, all negative (`docs/results/EXPLAIN_PERMUTATION.md`):

1. *More conditioning capacity* — prefix 4 → 16 vectors, `cond_dim` 64 → 256, deeper projection, 10% condition dropout. No effect.
2. *A differentiable contrastive objective* — each brief must be cheaper under its own condition than under its neighbours'. Unit-tested to reach the prefix parameters. In training the loss sat at $\log B$, the value for a model that

ignores its condition, and in one seed rose.

3. *2.5x the training budget* — 500 iterations, gap logged every 100:

iteration	NLL/token (plain)	gap	NLL/token (contrastive)	gap
100	0.662	+0.0015	0.669	+0.0012
200	0.544	+0.0027	0.542	-0.0004
300	0.757	-0.0097	0.533	-0.0019
400	1.000	+0.0000	0.515	+0.0041
500	0.524	-0.0229	0.674	-0.0209

NLL plateaus near 0.5 nats/token; the gap never leaves zero.

Not a plumbing bug. On the Qwen path the prefix receives a gradient of norm $1.1e5$ (against $4.6e3$ for the LoRA adapters), the NLL moves by 1.51 when the condition is zeroed, and prefix embeddings differ across rows. The model *can* use the condition; the optimisation settles on the marginal distribution over actions rather than the conditional, and greedy decoding then emits the modal brief every time.

The testbed could not have answered the question. The trained dar policy is effectively state-independent: across 64 states the action std is **1.2%** of the action magnitude, and on the brief's 0.05 quantisation grid those 64 states give **8 distinct action vectors**. Its briefs are nearly identical, its supervised NLL reaches 0.05 nats/token, and a constant brief is close to correct. On dar the permutation gap is zero no matter what the generator does. Every Explain number measured on dar was computed on a task with nothing to explain. NDWS — 1,013 distinct plans across 1,024 fires, across-fire intensity std 3.06x the mean — is the only testbed here where the question is posed.

What stands. The certificate machinery: coverage holds at 90% and 95% on 3 of 3 seeds on real data, and the floor it reports is honest about the generator it was given. Certifying a constant brief is not a defect of the certificate.

What to change. Report the gap (aligned minus shuffled), not F. Retire dar as an Explain testbed and calibrate any replacement for action diversity. Treat the fix as a design change — cross-attention conditioning, or a structured head for the (patch, amplitude) list with prose generated around it — not a tuning exercise.

Part IV — What survives, and what does not

Survives multi-seed scrutiny

- Return advantage over four safe-RL baselines **on dar** ($t = +3.84$ to $+17.95$)
- Real-sample efficiency: 9x on dar, 5x on rdf
- FNO » DeepONet » GNOT, on synthetic and real benchmark data
- Surrogate accuracy competitive with published PDEBench FNO error
- Resolution invariance to 128^2 (16,384 cells)
- Constraint violations cut $\sim 7\times$ by probe + margin; 13 of 15 seed-runs clean
- A conformal margin with a **stated failure rate**: at $\delta = 0.1$ the realised violation rate was 0% (dar) and 7.3% (rdf), both inside the bound
- Joint training is safe when data-anchored (surrogate improves 5–9x, decision unchanged) and destructive when not, on all three testbeds
- Assumption 1 ($L_G \leq 1$) after training
- The conformal certificate's coverage, on synthetic and real data
- The FNO's scope: operators for smooth fields, convolutions for sharp fronts (NDWS gate 0.12, U-Net > hybrid at $t = +12.5$)
- U-Net on real wildfire data at 97.5% of published AUC-PR

- Constrained planning on real fire data: +12.0 points over forecast-then-greedy (5 seeds, $t = 11.15$), 1.9–2.1x the heuristic's reduction at tight budgets
- That planning advantage **without** the imposed action physics: the gap widens to +13.6 points when only the learned fuel response is active
- The mechanism behind it: the advantage is insignificant at a one-day horizon (where greedy is the exact optimum, $t = 1.40$) and grows monotonically to +14.3 at five days, while the heuristic degrades

Does not survive

- Constraint satisfaction *as originally built* — 7.3% of evaluations violated
- Constraint satisfaction as a *guarantee* — the fix is a 7x reduction, with two excursions in 15 seed-runs
- A return edge on rdf — parity with PPO-Lagrangian at matched safety
- Joint training *beating* the disaggregated pipeline on return
- Equation 8 improving on the variance-only α (measured B^2 is too small)
- Proposition 1's infinite-horizon bound as a usable number (4,000x loose)
- The physics-informed loss improving asymptotic accuracy
- Adaptive α improving mean return (it improves the tail)
- The faithfulness objective contributing beyond supervised brief training
- PSPE perception beating a CNN or a frozen probe on reconstruction
- Any specific magnitude for transfer into the wave family
- **Any claim that a brief explains the decision it was generated for**
- `swe` as a planning testbed — not because the policy fails to act (it beats doing nothing by +0.0122, better than a hand-written damper) but because the entire span between doing nothing and the best achievable return is ~7% of the objective, so method differences sit inside it
- `dar` as an Explain testbed — its policy is state-independent

Not yet measured

- Flood testbed on real data; human study on real briefs
- Safe-RL baselines on `swe` (blocked on the `swe` task redesign)
- NDWS test-split numbers with focal loss
- Safe-RL baselines on `swe` (blocked on its redesign)

Part V — Measurement defects found and fixed

This project's most transferable output may be the list of ways its own measurements were wrong. Each was found by a check that should be standard.

defect	effect	how it was caught
Constraint read from one final evaluation snapshot	Identical runs scored "0% violation" or "100%" on timing luck	Multi-seed reruns disagreeing
Evaluation perturbed training through the global RNG	Baseline returns shifted -2.51 → -2.30	Forking the RNG changed the baseline
Optional-dependency tests skipped silently	A "tested" loader failed on first real use	<code>importorskip</code> audit
Truth solver unstable above 64 ²	NaN at every resolution above the training grid	Resolution sweep
Faithfulness REINFORCE had exactly zero gradient	Samples never parsed; every score equalled the parser fallback	Parse-rate diagnostic
Cost limits uncalibrated	The dual never activated; five unconstrained learners tying	Measuring do-nothing and greedy cost
$F = \exp(-KL)$ with KL summed over dimensions	Every arm scored exactly 0.0 at 64 actions while the reference scored 0.985	Running Explain on a large action space
Faithfulness never measured state-specificity	A constant brief scores as well as a correct one	Permutation control
dar's policy is state-independent	The Explain testbed had nothing to explain	Measuring action diversity across states
Conformal margin bounded the surrogate's error, not the policy's spread	Violations rose to 18.2% on rdf, worse than no margin at all	Running it on a testbed where the surrogate is accurate but the policy is variable
Baselines compared against their own initialisation	Four constrained-RL methods "failed structurally" at 22–38% of a 3% budget	Solving for the initial action the parameterisation implies
Testbed calibrated for a binding constraint but not a controllable objective	swe ranked planning methods inside a 7% band for the whole project	Measuring return separation, not just cost separation

Seven of these checks, their standing rules, and the three results they overturned *before* publication are collected in `docs/results/ROBUSTNESS_NOTES.md`.

The general lesson. Every metric that compares a generated artefact to a target should ship with a permutation control, and every testbed should be calibrated for the variation the metric needs before it is used. Without those, "the metric went up" and "the model learned the task" are indistinguishable — and in this project they were, for three rounds of scrutiny including a real-backbone replication and a conformal certificate.

Part VI — Infrastructure and reproduction

Protocol. 5 seeds by default, mean $\pm$ sample std, paired t -tests. 171 tests including regression tests for each measurement defect above. One command per sweep (`eval/run_seeds.py`), with aggregation by run or arm key.

Compute. TACC Vista (GH200). Jobs are packed — all seeds of one experiment concurrently on one GPU — because a single short job backfills into gaps a multi-node array never fits. A 5-seed sweep takes about 10 minutes against ~9 hours on one CPU. Time limits matter more than node counts: 25–30 minute jobs backfill within minutes when hour-long jobs wait hours.

Key scripts.

purpose	command
Multi-seed sweep	<code>python eval/run_seeds.py --seeds 0 1 2 3 4 --full</code>
Constraint fix / planner v2	<code>python eval/run_constraint_fix.py --sat-coef 1 --adv-floor 1e-2 --dual-ema 0.7 --ki 0.5 --margin-episode-std</code>
Joint vs disaggregated	<code>python eval/run_joint.py --beta 0.1</code>
NDWS surrogate	<code>python eval/run_ndws.py --epochs 18 --n-train 8000</code>
NDWS planning	<code>python eval/run_ndws_planning.py --horizon 3</code>
NDWS briefs + certificate	<code>python eval/run_ndws_explain.py --arms trained-in</code>
Permutation control	<code>python eval/run_explain_baselines.py --n-permutation 128</code>
Vista submission	<code>sbatch --export=ALL,EXPERIMENT=<name>[,TESTBED=,ARMS=,CW=] scripts/tacc/vista_packed.slurm</code>

Detailed analyses.

file	contents
<code>docs/results/PHASE1_ANALYSIS.md</code>	Joint training, Eq. 8, Lipschitz/Prop 1, faithfulness weight sweep, conformal on dar
<code>docs/results/PHASE3_ANALYSIS.md</code>	Constraint fix and joint training on swe and rdf
<code>docs/results/RDF_PLANNER_FIX.md</code>	The three-defect diagnosis, the dar regression, rdf safe-RL baselines
<code>docs/results/NDWS_PLANNING.md</code>	Constrained firebreak planning on observed fire data
<code>docs/results/NDWS_EXPLAIN.md</code>	Briefs and certificate on real fire plans
<code>docs/results/EXPLAIN_PERMUTATION.md</code>	The permutation control and three failed fixes
<code>docs/technical_report.md</code>	Full narrative report with method and literature
<code>docs/slides_experiments.md</code>	Slide-by-slide version of the experimental section
<code>docs/figures/</code>	Architecture, module and application diagrams

Part VII — Where this leaves the work

The defensible framing. Not "joint training of four modules beats disaggregated pipelines" — that was tested on three PDE families and is false. The framing the evidence supports is:

Learned surrogates make constrained planning cheap, but the planner inherits every error the surrogate makes, silently. Measure the real system periodically, plan against the measured error, and the violations mostly go away at no return cost and a fraction of the real-world samples. Where the decision is genuinely sequential, planning through the surrogate beats the operational heuristic on real data. And the parts that do not work — joint training, the faithfulness objective, the explanation module's conditioning — are reported with the measurements that show why.

Three contributions stand. Safety by measurement; sample efficiency; planning on observed data. Each is multi-seed, mechanism-explained, and reproduced across testbeds where applicable.

One is halved. The explanation module has a working certificate and an open conditioning problem, both measured.

The immediate next experiments, in order of value:

1. ~~NDWS test-split numbers~~ — **done**, and the surrogate exceeds the published figure (§Simulate). Remaining: numbers with focal loss, to close the $0.277 \rightarrow 0.284$ gap on the split the published figure uses.
2. ~~The swe task redesign~~ — **attempted and characterised** (§3b). The actuator-amplitude fix does not work on a stochastic policy; a new objective would be needed. Retired as a comparison testbed with a measured reason.
3. The former item 2 — more actuation authority until the do-nothing and greedy costs separate — or its retirement as a planning testbed.
4. The Explain redesign (structured head for the action list), which is a different module and possibly a different paper.

Done since the first draft of this report: the NDWS test-split result (§Simulate), the swe characterisation (§3b), the NDWS result extended to five seeds (§5), the crew-budget Pareto sweep (§5a), constrained-RL baselines on the same fires (§5a2), the horizon ablation at five seeds that shows the mechanism (§5a3), the fuel-only ablation (§5b), and a working conformal margin (§1). The robustness checks that overturned two results before they were reported are in `docs/results/ROBUSTNESS_NOTES.md`.